ModalIn: Decentralized P2P Micro-Lending with On-Chain Reputation Using Soulbound Tokens and Social Trust Primitives for Indonesian MSMEs

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***Abstract*—Indonesia's 64 million micro, small, and medium enterprises (MSMEs) contribute 61% of national GDP yet face a credit gap exceeding Rp 2,800 trillion annually due to the absence of verifiable credit history and physical collateral. Existing decentralized finance (DeFi) platforms fail this population because they require overcollateralization in crypto assets. We present ModalIn, an Ethereum-based peer-to-peer micro-lending decentralized application (DApp) that replaces physical collateral with a triple-layer on-chain reputation system. ModalIn issues each borrower a non-transferable Soulbound Token (SBT) as a persistent digital credit identity, organizes borrowers into tiered Credit Guilds that formalize Indonesia's collective accountability culture, and enables ETH-staked peer vouching as social guarantee. A composite reputation score weighted 50% payment history, 30% vouch score, and 20% oracle attestation drives an algorithmic annual percentage rate (APR) model bounded between 6% and 36% in alignment with Indonesian Financial Services Authority (OJK) reference bounds. The system comprises six interacting Solidity smart contracts deployed on the Ethereum Sepolia testnet, validated by 29 passing test scenarios. Results show that a Gold-tier, high-reputation borrower achieves an APR as low as 6% compared to 17% for an unscored borrower, a reduction of up to 65%, and a threefold improvement over typical informal moneylender rates.**

***Index Terms*—blockchain, soulbound token, decentralized finance, micro-lending, credit scoring, peer-to-peer lending, smart contract, MSME**

I. Introduction

Micro, small, and medium enterprises (MSMEs) form the backbone of Indonesia's economy. With over 64 million units, they contribute approximately 61% of national gross domestic product and absorb 97% of the domestic workforce [4]. Despite this economic significance, more than 40% of Indonesian MSMEs lack access to formal banking credit, creating a financing gap estimated at over Rp 2,800 trillion per year [4]. The root causes are structural: conventional credit scoring systems require documented repayment histories and physical assets as collateral, prerequisites that most informal-sector MSME operators cannot meet [3].

Unable to access formal channels, these businesses resort to informal moneylenders (*rentenir*) whose annualized interest rates commonly reach 20%–60% [9]. This creates a vicious cycle: prohibitive interest burden prevents capital accumulation, which in turn prevents the formation of credit history needed to qualify for cheaper formal loans.

Blockchain technology offers a structural solution by resolving information asymmetry between lenders and borrowers without a trusted intermediary [4]. The immutability of on-chain records ensures that repayment history cannot be manipulated, providing lenders with mathematical certainty. The Soulbound Token (SBT), a non-transferable, wallet-bound digital artifact representing commitments and credentials [1], enables a persistent, unforgeable credit identity that follows a borrower across all transactions on the network.

Critically, Indonesian informal credit culture already embeds the primitives needed for reputation-based lending. The *arisan* (rotating savings group), *koperasi* (cooperative), and *gotong royong* (mutual aid) traditions demonstrate that collective accountability and social guarantee are culturally validated mechanisms for reducing default risk. What has been missing is infrastructure that encodes these social trust mechanisms into verifiable, immutable on-chain data.

We present **ModalIn**, a decentralized P2P micro-lending DApp that translates these cultural trust primitives into smart contract logic on the Ethereum Virtual Machine (EVM). The contributions of this paper are:

1) A **triple-layer on-chain reputation model** combining payment history, ETH-staked peer vouching, and off-chain oracle attestation into a composite credit score on a 0–1000 scale, with formal weight assignment and a time-based reputation decay mechanism.

2) An **algorithmic interest rate model** computing APR directly from on-chain reputation and group tier, with OJK-referenced bounds (6%–36%), eliminating manual underwriting.
3) A **Credit Guild mechanism** (Bronze/Silver/Gold tier) formalizing *gotong royong* collective accountability as a smart contract primitive with group-level score propagation.
4) A **full working DApp**: six Solidity contracts (Hardhat 3.1.12, OpenZeppelin v5.6), a React 19 frontend with ethers.js v6 integration, deployed on the Ethereum Sepolia testnet, and validated by 29 automated test scenarios.

II. Related Work

A. Overcollateralized DeFi Lending

Dominant DeFi lending protocols such as Aave and Compound use overcollateral- ization, requiring borrowers to lock crypto assets worth 120%–150% of the loan value [7]. While this protects lenders, it is fundamentally inaccessible to MSMEs who hold no crypto assets. Goldfinch [14] partially addresses this by introducing off-chain trust anchors, but still relies on institutional backers rather than individual on-chain reputation. ModalIn departs from this paradigm entirely: no crypto collateral is required, and reputation is the sole collateral.

B. Traditional Group Lending

The Grameen Bank model [13] demonstrated that group-based social pressure can reduce default rates to below 2% in microcredit populations with no physical collateral. However, Grameen's model is manual, centralized, and non-auditable at global scale. ModalIn encodes the same social accountability mechanism (collective responsibility, tiered trust, and social guarantee) as immutable smart contract logic.

C. Blockchain for SME Finance

Kumar et al. [4] conducted a systematic review of blockchain-based SME finance and found strong evidence that distributed ledgers reduce information asymmetry and transaction costs, but their work remains a literature review without a system architecture. Kanga et al. [3] examined blockchain-based microcredit platforms in developing countries and identified on-chain identity as the critical missing component for financial inclusion. Ante et al. [7] analyzed blockchain as a solution for P2P lending fraud, concluding that immutability addresses the core trust problem but offered no reputation scoring mechanism. Nanayakkara et al. [5] proposed swarm learning for P2P credit scoring, but their approach relies on off-chain machine learning rather than on-chain identity primitives.

D. On-Chain Reputation and Soulbound Tokens

Weyl, Ohlhaver, and Buterin [1] introduced Decentralized Society (DeSoc) and SBTs as non-transferable tokens encoding social relationships and credentials. EIP-5114 [12] standardizes the Soulbound Badge interface. ModalIn operationalizes these concepts specifically for credit scoring with a weighted multi-dimensional formula. Table I positions ModalIn against the most closely related systems.

TABLE I
COMPARISON OF RELATED SYSTEMS

System	Mechanism	Key Limitation	Collateral
Aave/Compound	Overcollateral	Excludes unbanked	Crypto
Grameen Bank	Group lending	Manual, centralized	Social
Goldfinch	Off-chain trust	Institutional backers	Off-chain
Kumar et al.	Review only	No architecture	N/A
ModalIn	**SBT+Guild+Vouch**	**ETH denomination**	**Reputation**

III. Proposed Method

A. Problem Formalization

Given a borrower b with no prior formal credit record, we seek to compute a verifiable trust score $S(b) \in [0, 1000]$ derived solely from on-chain behavior, and use it to determine a fair lending APR:

$$\text{APR}(b) = \text{BaseRate} + \text{GroupPremium}(b) - \text{RepDiscount}(b) \tag{1}$$

where all terms are in basis points (1 bps = 0.01%), and APR is clamped to $[\text{APR}_{\text{min}}, \text{APR}_{\text{max}}] = [600, 3600]$ bps, corresponding to 6%–36% annually in alignment with OJK reference bounds.

B. Triple-Layer Reputation Model

The composite credit score aggregates three orthogonal trust signals:

$$S(b) = \frac{w_p \cdot P(b) + w_v \cdot V(b) + w_a \cdot A(b)}{100} \tag{2}$$

where $P(b) \in [0, 1000]$ is the **payment score** computed as $10r$ with $r = (\text{loans repaid}/\text{loans taken}) \times 100$; $V(b) \in [0, 1000]$ is the **vouch score** from ETH-staked peer guarantees computed by `VouchRegistry`; $A(b) \in [0, 1000]$ is the **attestation score** from an authorized oracle (e.g., e-commerce transaction data); and weights $w_p = 50$, $w_v = 30$, $w_a = 20$ sum to 100.

The weight assignment reflects the hierarchy of manipulation resistance: payment history is fully on-chain and hardest to forge; vouching requires real ETH at stake; attestation carries oracle centralization risk, hence the lowest weight.

C. Reputation Decay

To prevent stale high scores from being exploited, a time-based decay applies a penalty of 10 points per 180-day period of inactivity:

$$S_{\text{final}}(b) = \max\left(0,\ S(b) - 10 \left\lfloor \frac{t_{\text{now}} - t_{\text{last}}}{T_{\text{decay}}} \right\rfloor\right) \tag{3}$$

where $T_{\text{decay}} = 180$ days and t_{last} is the timestamp of the last on-chain update stored in the borrower's SBT profile.

TABLE II
INDONESIAN SOCIAL TRUST TO SMART CONTRACT MAPPING

Indonesian Tradition	**ModalIn Mechanism**	**Contract**
Credit bureau / bank history	Payment score via SBT	`SoulboundToken`
Koperasi / Arisan	Credit Guild (Bronze/Silver/Gold)	`GuildSBT`
Jaminan / guarantee	Peer vouch + ETH slash	`VouchRegistry`
Merchant sales data	Attestation oracle	`ReputationEngine`
Loan escrow	P2P lifecycle	`LoanEscrow`

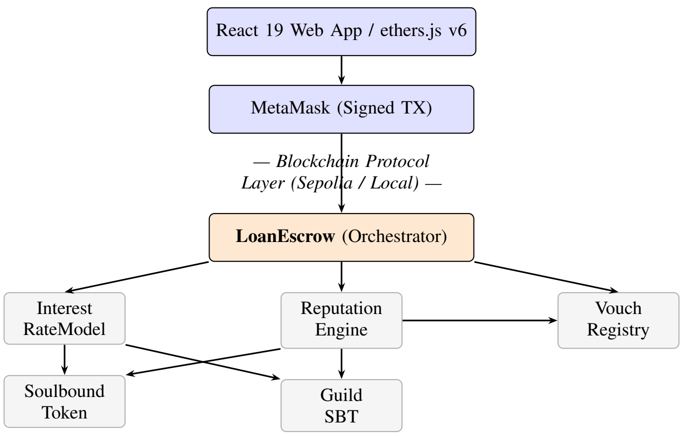

Fig. 1. ModalIn two-layer architecture. The frontend communicates exclusively through signed transactions; all loan and reputation state is stored on-chain.

D. Social Trust Mapping

A central design insight of ModalIn is the direct mapping of Indonesian social trust practices to on-chain primitives, as shown in Table II.

IV. SYSTEM DESIGN

A. Architecture Overview

ModalIn adopts a strict separation-of-concerns architecture across two layers, as shown in Fig. 1. The **Frontend Layer** is a React 19 web application that communicates through ethers.js v6 and a MetaMask wallet extension. All state-changing interactions are submitted as cryptographically signed transactions; no centralized database stores any loan or reputation state. The **Blockchain Protocol Layer** consists of six Solidity contracts with a defined dependency hierarchy: `LoanEscrow` is the primary orchestrator, delegating APR computation to `InterestRateModel` and reputation updates to `ReputationEngine`, which reads from and writes to `SoulboundToken`, `GuildSBT`, and `VouchRegistry`.

B. Smart Contract Design

Table III summarizes the six contracts and their responsibilities. The `SoulboundToken` stores each borrower's `CreditProfile` struct (encompassing loan counts, cumulative amounts, reputation score, and last-updated timestamp)

TABLE III
SMART CONTRACT SUMMARY

Contract	**Key Responsibility**
`SoulboundToken`	Non-transferable ERC-5114-inspired credit identity; stores `CreditProfile`; starting score 500; `transfer()` permanently reverts.
`GuildSBT`	Credit group (2–10 members); tier Bronze/Silver/Gold from collective score; tier updates on `recalculateScore()`.
`VouchRegistry`	ETH-staked peer vouches; stake-weighted vouch score; slashes on default.
`ReputationEngine`	Aggregates $S(b)$ per (2); applies decay per (3); propagates to guild average.
`InterestRateModel`	APR per (1): Bronze +800 bps, Silver +400 bps, Gold +0 bps; reputation discount up to -600 bps.
`LoanEscrow`	Loan states: Requested $\rightarrow$ Active $\rightarrow$ Repaid/Defaulted; holds ETH in escrow; triggers reputation update on resolution.

TABLE IV
TECHNOLOGY STACK

Component	**Version**	**Role**
Solidity	0.8.20	Smart contract language
Hardhat	3.1.12	EVM dev environment
OpenZeppelin	v5.6.1	Ownable, ReentrancyGuard
ethers.js	v6.16.0	Frontend–contract bridge
React	19	Frontend framework
TypeScript	v5.8.2	Frontend type safety
Vite	v6.2.0	Build toolchain
TailwindCSS	v4.1.14	UI styling
Mocha / Chai	11.x / 6.x	Test runner / assertions
Node.js	v24.11.1	Runtime environment

and starts new borrowers at a neutral score of 500. The `GuildSBT` assigns tier based on collective average score: Bronze ($S < 650$), Silver ($650 \leq S < 800$), Gold ($S \geq 800$). The `VouchRegistry` records ETH-staked peer guarantees and slashes stakes proportionally upon borrower default.

C. Access Control and Security Design

Sensitive state-writing functions (`updateReputation`, `recordLoan`, `recordRepayment`) are protected by an `authorizedUpdaters` whitelist pattern rather than role-based access control, keeping gas overhead low while preventing unauthorized score manipulation. All ETH-moving functions (`fundLoan`, `repayLoan`, `withdrawFunds`) are guarded with OpenZeppelin's `ReentrancyGuard`. `LoanEscrow` additionally follows the Checks-Effects-Interactions (CEI) pattern: all state updates precede external ETH transfers, eliminating reentrancy vectors by design.

V. IMPLEMENTATION

A. Technology Stack

Table IV lists the exact component versions used in ModalIn.

B. Deployment Strategy

Two deployment environments are supported. For local development, `hardhat node` instantiates a simulated EVM (Chain ID 31337) with controllable block timestamps, enabling testing of deadline and decay scenarios. The deploy script deploys all six contracts in dependency order and injects their addresses into `modalin-frontend/src/abis/contract-addresses.json`, making the frontend environment-agnostic. For public validation, contracts were deployed to the **Ethereum Sepolia testnet** (Proof of Stake) using `deploy:sepolia`, demonstrating viability with real consensus and block finality.

C. Key Implementation Decisions

Basis-point APR arithmetic. Solidity does not support floating-point. All rates are stored in integer basis points and interest is computed as:

$$\text{interest} = \frac{\text{principal} \times \text{APR(bps)} \times \text{days}}{365 \times 10000}$$

Self-registration. `SoulboundToken.selfRegister()` allows any wallet to issue its own SBT on first connection, eliminating admin-gated onboarding and reducing entry friction for MSME users.

Guild size cap. Membership is capped at 10 per group to bound the iteration in `_updateGroupScore()`, preventing out-of-gas exceptions during collective score recalculation on the EVM.

Custom Solidity errors. All revert conditions use typed `error` declarations (e.g., `AlreadyHasSBT`, `NotAuthorizedOracle`) instead of `require` string messages, reducing deployment and call gas costs.

VI. Testing and Evaluation

A. Test Suite

ModalIn employs a three-level testing strategy: (1) **Unit Testing**: each contract function tested in isolation; (2) **Integration Testing**: cross-contract interactions (e.g., `LoanEscrow` invoking `ReputationEngine` on repayment); and (3) **Manual UI Testing**: end-to-end simulation via the frontend and MetaMask to validate the borrower and lender user journeys. The automated suite uses Hardhat's `evm_increaseTime` RPC call to manipulate block timestamps, enabling deadline and decay scenarios without waiting for real time to elapse.

All 29 automated test scenarios pass under Mocha v11 and Chai v6. Table V summarizes results by contract grouping.

TABLE V
Test Suite Results (29/29 Passing)

Contract / Scenario Group	Tests	Result
SoulboundToken	6	6 ✓
GuildSBT	5	5 ✓
VouchRegistry	4	4 ✓
ReputationEngine	4	4 ✓
InterestRateModel	3	3 ✓
LoanEscrow: lifecycle	6	6 ✓
LoanEscrow: default/slash	1	1 ✓
Total	**29**	**29 ✓**

B. Functional and Performance Testing

Table VI presents the results of manual UI testing for the seven core user-facing features. All scenarios pass end-to-end using the React frontend and MetaMask on a local Hardhat node.

TABLE VI
Functional Test Results (Manual UI)

Feature	Expected Outcome	Result
Identity Registration	`selfRegister` issues SBT, score = 500	✓
Group Creation	Bronze group formed; creator auto-enrolled	✓
Social Vouch	ETH locked; vouch count increments	✓
Loan Request	Status = 0; APR computed automatically	✓
Loan Funding	Status = 2; ETH auto-transferred to borrower	✓
Loan Repayment	Status = 3; reputation score rises	✓
Default & Slash	Status = 4; voucher stakes slashed	✓

A UAT-phase race condition was identified and fixed: on first MetaMask connection the frontend called `selfRegister` twice in rapid succession, triggering `AlreadyHasSBT` on the second call. The fix silently swallows `AlreadyHasSBT` via a try-catch in the frontend connection handler.

Performance. On the local Hardhat node, transactions are confirmed instantaneously (<1 second). On the Ethereum Sepolia testnet, finality follows the Ethereum block time of approximately 12 seconds per transaction. The guild-score loop is bounded to a maximum of 10 iterations, ensuring `_updateGroupScore` never exceeds the block gas limit.

C. APR and Reputation Scenario Analysis

Table VII presents APR outcomes for representative borrower profiles computed from the deployed `InterestRateModel` contract.

These results are derived analytically from (1):

- **New borrower, Bronze**: $1200 + 800 - (500 \times 600/1000) = 1700$ bps (17%).
- **High-rep, Gold**: $1200+0-(850\times600/1000) = 690$ bps, clamped to 600 bps (6%).

These values are confirmed by test assertions: *"returns APR above 1500 bps for Bronze/low-rep borrower"* and *"returns APR at or below 700 bps for Gold/high-rep borrower"*, both of which pass in the suite.

The maximum APR reduction, from 17% (unscored, no group) to 6% (Gold tier, top reputation), represents a **65% reduction in borrowing cost**. Relative to informal moneylenders (20%–60% APR) [9], a Gold-tier ModalIn borrower achieves a

TABLE VII
APR OUTCOMES BY BORROWER PROFILE

Borrower Profile	Score	Tier	APR
New borrower, no group	500	Bronze	17.0%
3 repayments, Bronze group	700	Bronze	15.8%
New borrower, Silver group	500	Silver	13.0%
Active repayer, Gold group	800	Gold	7.2%
High-rep, Gold group	850	Gold	$\leq$7%[a]

[a]Clamped to minimum APR floor (600 bps = 6%).

minimum threefold improvement in cost of capital, validating the core hypothesis that on-chain reputation can substitute for physical collateral.

D. Security Validation

The system was evaluated against the three most common smart contract attack vectors:

Reentrancy (PASS). All ETH-moving functions use OpenZeppelin's `ReentrancyGuard` modifier. `LoanEscrow.repayLoan()` additionally follows the Checks-Effects-Interactions (CEI) pattern: status is set to `Repaid` before any ETH transfer, so a re-entrant call finds `loan.status != Active` and reverts immediately.

Double Spending (PASS). `repayLoan()` checks `loan.status == Active` as its first guard. A loan in `Repaid` or `Defaulted` state cannot be repaid a second time.

Access Control (PASS). The `onlyAuthorized` modifier restricts score-writing functions to the whitelist in `authorizedUpdaters`. The soulbound guarantee is enforced by an unconditional `revert` in `transfer()`, preventing any secondary-market sale of credit identities. Additional test assertions confirm: reverts on second SBT issuance, reverts on self-vouching, and reverts when reputation weights do not sum to 100.

E. Discussion

The 65% APR reduction demonstrates that on-chain reputation can serve as a meaningful substitute for physical collateral. The guild mechanism provides a second incentive channel: joining a Gold-tier guild reduces APR from 17% to 9% for a borrower with no repayment history (score 500), creating a strong economic incentive to participate in social trust networks. The voucher-slashing mechanism, confirmed by the default test scenario, creates the same social pressure as Grameen's group model but with transparent, immutable on-chain enforcement.

Table VIII compares ModalIn against conventional P2P lending systems across four operational dimensions.

VII. CONCLUSION

This paper presented ModalIn, a decentralized P2P micro-lending DApp that addresses the credit gap faced by Indonesian MSMEs by replacing physical collateral with a triple-layer on-chain reputation system. The system formalizes Indonesia's social trust culture (*arisan*, *koperasi*, and *gotong royong*) as composable smart contract primitives: Soulbound Tokens for persistent credit identity, Credit Guilds for collective accountability, and ETH-staked peer vouching as social guarantee. An algorithmic interest rate model driven by the composite reputation score provides dynamic APR pricing within OJK-referenced regulatory bounds.

TABLE VIII
MODALIN VS. CONVENTIONAL P2P LENDING

Parameter	Conventional P2P / Bank	ModalIn (Blockchain)
Collateral	Physical asset or BPKB certificate required	None. Social vouch and on-chain reputation only.
Interest Rate	Set manually by analysts (subjective, opaque)	Algorithmic and transparent, derived from on-chain score.
Identity Ownership	Held by central bank or institution server	Sovereign. SBT is wallet-bound and user-controlled.
Disbursement	Days (human approval process)	Instant. ETH released automatically when escrow target is met.

All 29 test scenarios pass, validating the complete loan lifecycle including the default and voucher-slashing path. Evaluation of the APR model demonstrates up to a 65% reduction in borrowing cost for high-reputation borrowers compared to unscored entrants, and a minimum threefold improvement over informal moneylender rates, confirming that reputation can function as collateral.

Current limitations include: (1) Ethereum L1 transaction fees (gas) make loans below approximately Rp 100,000 economically irrational, as gas cost would exceed interest income; (2) the MetaMask wallet interface presents a significant adoption barrier for grassroots MSME users unfamiliar with private-key management; (3) the attestation oracle is a single trusted address, introducing centralization risk for off-chain data; (4) the undercollateralized model means lenders bear residual loss if borrowers accept reputational destruction over repayment; (5) no formal smart contract security audit has been conducted.

Future work is directed along four axes: (1) **Layer-2 Migration** to Polygon or Arbitrum to reduce gas costs and make sub-Rp 100,000 micro-loans economically viable; (2) **EAS Oracle Integration** using the Ethereum Attestation Service to pull private e-commerce sales records (Shopee, Tokopedia) as trustless off-chain attestation without a centralized oracle; (3) **ERC-4337 Account Abstraction** to allow MSME users to authenticate via email or phone number, eliminating the MetaMask seed-phrase barrier; and (4) **DeFi Insurance / Treasury Reserve Fund** to socialize lender losses in mass-default scenarios.

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